

Sabrina Sousa

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Experience

Technical Consultant | Contax Inc.

May 2023 – Aug 2024

- Resolved client needs with custom solutions for enterprise resource planning (ERP) system called SAP
- Created reports using **ABAP** and **SQL** for database queries and updates, order automation and GUIs
- Programmed cross-platform web applications using **JavaScript**, **XML** and **SAPUI5** (HTML5 framework)

Mechatronics Technologist | NRC Canmet Materials

May 2022 – Aug 2022

- 3D modelled designs** and **simulations** to build corrosion and material testing **automation** projects
- Designed and **3D printed** a prototype of a sealing chamber with a **camshaft** and **servo** mechanism
- Programmed waypoints for a **robotic arm** to complete **automated** cycles of metal alloy casting

Research Assistant | McMaster University

May 2021 – Aug 2021

- Developed and tested **Python** code to design hexagonal tiling of a prototype helmet for **optimal energy** absorption to study the impact of deformations on the helmet's ability to absorb impacts
- Named as a **coauthor** for the research presented at the CMSC conference

Education

Master of Cybersecurity and Threat Intelligence | University of Guelph

Jan 2026 – Present

- Courses: Cryptography, Threat Intelligence & Risk Analysis, Digital Forensics & Incident Response
- Awards: CEPS Dean's Graduate Excellence Entrance Scholarships (DGEES)

Bachelor of Engineering, Mechatronics Co-op | McMaster University

Sep 2020 – Apr 2025

- Cumulative GPA: 3.7 | Applicable Courses: Networks & Security (Grade: A), Software Dev (Grade: A)
- Awards: President's Entrance Scholarship, Dundas Scholarship, McMaster's Summer Research Award

Projects

Advanced Penetration Testing Custom Labs

May 2026 – Present

- Compromised **10+ machines** through vulnerability exploitation, privilege escalation, and remote exploitation via **Metasploit**, used **NSE** for fuzzing and cracked hashes with **Hashcat/John the Ripper**

Forensic Image Analysis Case Study

Apr 2026 – May 2026

- Conducted forensic analysis of an Android device image using **Autopsy** to recover evidence of a **phishing attack** and **credential theft**, to then produce a report that corroborated the prosecution's case

AI Malware Classification Model

Jan 2026 – Apr 2026

- Developed **classical machine learning** and **deep convolutional neural network** models to classify malware across **6 families** using opcode sequence analysis, evaluating tradeoffs in accuracy across models

Cryptography Encoder Application

Aug 2025 – Dec 2025

- Developed a **Java** application implementing cryptographic algorithms for secure message encryption
- Designed an intuitive, user-friendly GUI in **Java Swing**, focusing on usability and visual clarity

EyeCan Capstone Project

Sep 2024 – Apr 2025

- Designed a **real-time** heat detection system for visually impaired users to avoid hazards in the kitchen
- Utilized thermal camera **image processing** and **OpenCV** in **Python** to detect hazards and track a user
- Established **BLE communication** between a wearable **ESP32** device using **NimBLE** stack in **C** and a **Raspberry Pi** camera module with **Bleak Client** integrated in **Python** for reliable user feedback

Skills

Software and Programming: Python, C, C++, Java, JavaScript, SQL, XML, JSON, HTML5, Git, AWS, ABAP, Linux, Assembly, MATLAB, R, Verilog, JIRA, Wireshark, Autopsy, OpenVAS, Nessus, and Metasploit

Hardware: STM32, Raspberry Pi, Arduino, Arm, 3D Printing, Oscilloscope, and Soldering

Extracurriculars

Computer Chapter Vice Chair | McMaster IEEE Student Branch

Jan 2022 – Apr 2023

- Led a team of 15 students working on a remote-controlled car project using a **Raspberry Pi** with **lane detection** using **Python OpenCV** and **brushed DC motors** controlled by a connected **Arduino** in **C**